

Deep Generative Models

Lecture 3

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AI Masters

2026, Spring

Recap of Previous Lecture

Jacobian Matrix

Let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x}), \quad \mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

Change of Variables Theorem (CoV)

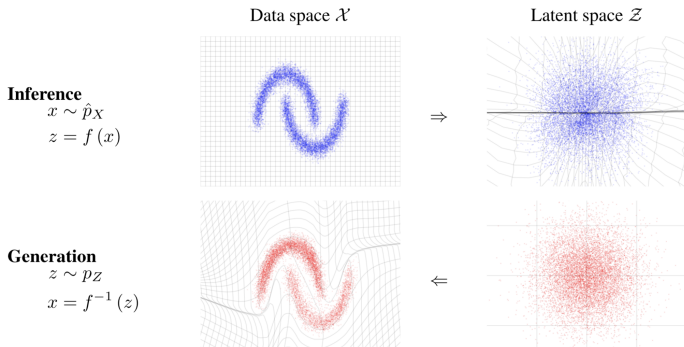
Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^1 -diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$
$$p(\mathbf{z}) = p(\mathbf{x}) |\det(\mathbf{J}_{\mathbf{f}^{-1}})| = p(\mathbf{x}) \left| \det \left(\frac{\partial \mathbf{x}}{\partial \mathbf{z}} \right) \right| = p(\mathbf{f}^{-1}(\mathbf{z})) \left| \det \left(\frac{\partial \mathbf{f}^{-1}(\mathbf{z})}{\partial \mathbf{z}} \right) \right|$$

Recap of Previous Lecture

Definition

A normalizing flow is a C^1 -diffeomorphism that transforms data $\mathbf{x}$ to noise $\mathbf{z}$.



Log-Likelihood

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^K \log |\det(\mathbf{J}_{\mathbf{f}_k})|$$

Recap of Previous Lecture

Flow Log-Likelihood

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_{\theta}(\mathbf{x})) + \log |\det(\mathbf{J}_{\mathbf{f}})|$$

One significant challenge is efficiently computing the Jacobian determinant.

Linear Flows

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \theta = \mathbf{W}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}^{\top}$$

- ▶ LU Decomposition:

$$\mathbf{W} = \mathbf{P}\mathbf{L}\mathbf{U}.$$

- ▶ QR Decomposition:

$$\mathbf{W} = \mathbf{Q}\mathbf{R}.$$

Decomposition is performed only once during initialization. Then the decomposed matrices (**P**, **L**, **U** or **Q**, **R**) are optimized.

Recap of Previous Lecture

Consider an autoregressive model:

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}), \quad p_{\theta}(x_j | \mathbf{x}_{1:j-1}) = \mathcal{N}(\mu_{j,\theta}(\mathbf{x}_{1:j-1}), \sigma_{j,\theta}^2(\mathbf{x}_{1:j-1})).$$

Gaussian Autoregressive Normalizing Flow

$$\mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z}) \quad \Rightarrow \quad x_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot z_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}).$$

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) \quad \Rightarrow \quad z_j = (x_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}.$$

- ▶ This transformation is both C^1 -**diffeomorphism**, mapping $p(\mathbf{z})$ to $p_{\theta}(\mathbf{x})$.
- ▶ The Jacobian matrix for this transformation is triangular.

The generative function $\mathbf{f}_{\theta}^{-1}(\mathbf{z})$ is **sequential**, while the inference function $\mathbf{f}_{\theta}(\mathbf{x})$ is **not sequential**.

Recap of Previous Lecture

Let us partition $\mathbf{x}$ and $\mathbf{z}$ into two groups:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}].$$

Coupling Layer

$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1; \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \sigma_{\theta}(\mathbf{z}_1) + \mu_{\theta}(\mathbf{z}_1). \end{cases} \quad \begin{cases} \mathbf{z}_1 = \mathbf{x}_1; \\ \mathbf{z}_2 = (\mathbf{x}_2 - \mu_{\theta}(\mathbf{x}_1)) \odot \frac{1}{\sigma_{\theta}(\mathbf{x}_1)}. \end{cases}$$

Both density estimation and sampling require just a single pass!

Jacobian

$$\det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & \mathbf{0}_{d \times (m-d)} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix} = \prod_{j=1}^{m-d} \frac{1}{\sigma_{j,\theta}(\mathbf{x}_1)}.$$

A coupling layer is a special instance of an gaussian autoregressive normalizing flow.

Recap of Previous Lecture

Maximum Likelihood Estimation (MLE) Problem

$$\theta^* = \arg \max_{\theta} p_{\theta}(\mathbf{X}) = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

The distribution $p_{\theta}(\mathbf{x})$ should be highly complex (just like the true data distribution $p_{\text{data}}(\mathbf{x})$).

Latent Variable

Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\theta}(\mathbf{x}, \mathbf{z}) = p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\theta}(\mathbf{x}, \mathbf{z}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

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$$p_{\theta}(\mathbf{x}) = \int p_{\theta}(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z}) d\mathbf{z}.$$

- ▶ $p(\mathbf{z})$ is prior distribution over the latent variable.
- ▶ $p_{\theta}(\mathbf{x}|\mathbf{z})$ is **decoder** distribution.
- ▶ Both $p_{\theta}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ are usually much simpler than $p_{\theta}(\mathbf{x})$.

Outline

1. Variational Evidence Lower Bound (ELBO)
2. Amortized Inference
3. ELBO Gradients, Reparametrization Trick
4. Variational Autoencoder (VAE)

Outline

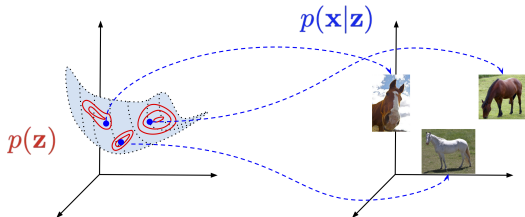
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MLE for LVM

$$\sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i | \mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \rightarrow \max_{\theta}.$$

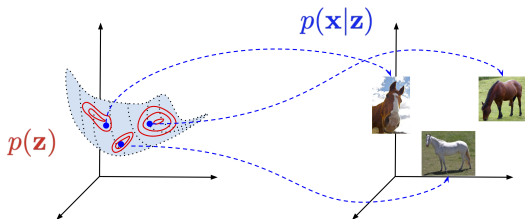
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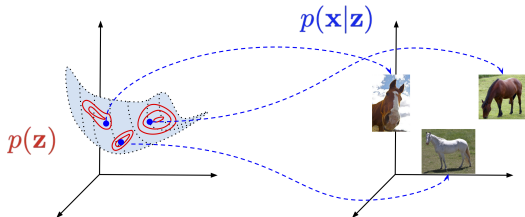
Naive Monte Carlo Estimation

$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k),$$

where $\mathbf{z}_k \sim p(\mathbf{z})$.

MLE for LVM

$$\sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i | \mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i \rightarrow \max_{\theta}.$$



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$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k),$$

where $\mathbf{z}_k \sim p(\mathbf{z})$.

Challenge: As the dimensionality of $\mathbf{z}$ increases, the number of samples needed to adequately cover the latent space grows exponentially.

ELBO Derivation I

Bayes theorem

$$p_{\theta}(\mathbf{z}|\mathbf{x}) = \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{x})} \quad \Rightarrow \quad p_{\theta}(\mathbf{x}) = \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})}$$

ELBO Derivation I

Bayes theorem

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Log likelihood

$$\log p_{\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z}$$

ELBO Derivation I

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Log likelihood

$$\log p_{\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} = \mathbb{E}_q \log \left[\frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} \right]$$

ELBO Derivation I

Bayes theorem

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Log likelihood

$$\begin{aligned} \log p_{\theta}(\mathbf{x}) &= \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}) d\mathbf{z} = \mathbb{E}_q \log \left[\frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})} \right] = \\ &= \mathbb{E}_q \log \left[\frac{p_{\theta}(\mathbf{x}, \mathbf{z}) q(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x}) q(\mathbf{z})} \right] \end{aligned}$$

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- ▶ Here, $q(\mathbf{z})$ is any distribution such that $\int q(\mathbf{z}) d\mathbf{z} = 1$.
- ▶ We assume that $\text{supp}(q(\mathbf{z})) = \text{supp}(p_{\theta}(\mathbf{z}|\mathbf{x})) = \mathbb{R}^d$.

ELBO Derivation II

Variational Decomposition

$$\log p_{\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} + \mathbb{E}_q \log \frac{q(\mathbf{z})}{p_{\theta}(\mathbf{z}|\mathbf{x})}$$

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Variational Evidence Lower Bound (ELBO)

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} \leq \log p_{\theta}(\mathbf{x})$$

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- This inequality holds for any choice of $q(\mathbf{z})$.

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- ▶ This inequality holds for any choice of $q(\mathbf{z})$.
- ▶ Instead of maximizing the likelihood, maximize the ELBO:

$$\max_{\theta} p_{\theta}(\mathbf{x}) \quad \rightarrow \quad \max_{q,\theta} \mathcal{L}_{q,\theta}(\mathbf{x})$$

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- ▶ Distribution $q(\mathbf{z})$ is treated as **variational** parameter.

Variational Evidence Lower Bound (ELBO)

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z}) \| p_{\theta}(\mathbf{z}|\mathbf{x}))$$

What is the optimal distribution $q^*(\mathbf{z})$ given fixed θ^* ?

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Here we got the intuition about variational distribution $q(\mathbf{z})$ – it estimates the posterior $p_{\theta^*}(\mathbf{z}|\mathbf{x})$.

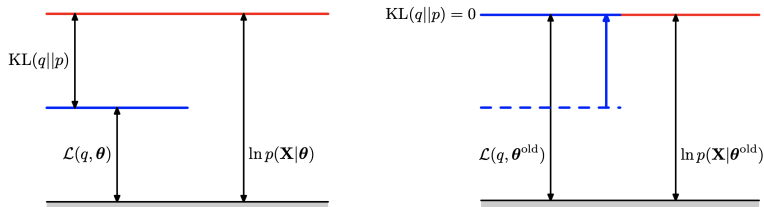
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Log-Likelihood Decomposition

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Variational Evidence Lower Bound (ELBO)

$$\begin{aligned}\mathcal{L}_{q,\theta}(\mathbf{x}) &= \int q(\mathbf{z}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \int q(\mathbf{z}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} + \int q(\mathbf{z}) \log \frac{p(\mathbf{z})}{q(\mathbf{z})} d\mathbf{z} \\ &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))\end{aligned}$$

Log-Likelihood Decomposition

$$\begin{aligned}\log p_{\theta}(\mathbf{x}) &= \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})) = \\ &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).\end{aligned}$$

- What do we get if we drop $D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$ term from ELBO?

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- ▶ What do we get if we drop $D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$ term from ELBO?
- ▶ What do we have in the term $\mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z})$ in the case of Normal distribution?

Outline

1. Variational Evidence Lower Bound (ELBO)
2. Amortized Inference
3. ELBO Gradients, Reparametrization Trick
4. Variational Autoencoder (VAE)

Parametric Variable Posterior

Variational Posterior

$$q(\mathbf{z}) = \arg \max_q \mathcal{L}_{q, \theta^*}(\mathbf{x}) = \arg \min_q D_{\text{KL}}(q \| p) = p_{\theta^*}(\mathbf{z} | \mathbf{x}).$$

Parametric Variable Posterior

Variational Posterior

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- ▶ $p_{\theta^*}(\mathbf{z} | \mathbf{x})$ may be **intractable**;
- ▶ $q(\mathbf{z})$ is individual for each data point $\mathbf{x}$.

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Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_\phi(\mathbf{z} | \mathbf{x})$, **conditioned on data $\mathbf{x}$** and **parameterized by ϕ** .

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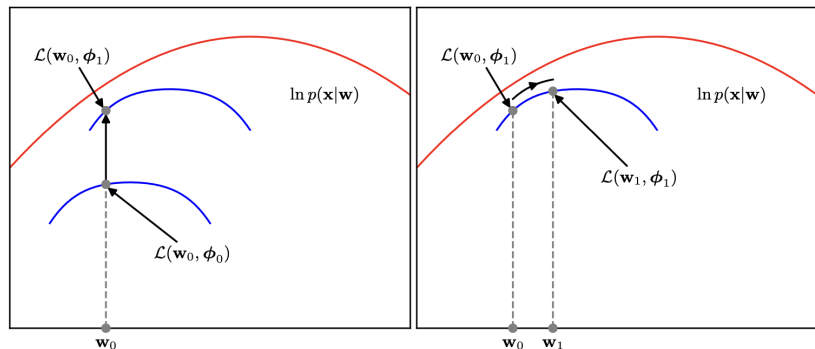
Gradient Update

$$\begin{bmatrix} \phi_k \\ \theta_k \end{bmatrix} = \left[\begin{bmatrix} \phi_{k-1} + \eta \cdot \nabla_\phi \mathcal{L}_{\phi, \theta}(\mathbf{x}) \\ \theta_{k-1} + \eta \cdot \nabla_\theta \mathcal{L}_{\phi, \theta}(\mathbf{x}) \end{bmatrix} \right]_{(\phi_{k-1}, \theta_{k-1})}$$

ELBO Optimization

Gradient Update

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ELBO Optimization

ELBO

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{\phi, \theta}(\mathbf{x}) + D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p_{\theta}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{\phi, \theta}(\mathbf{x}).$$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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- ▶ ϕ denotes the parameters of the variational posterior $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- ▶ θ represents the parameters of the generative model $p_{\theta}(\mathbf{x}|\mathbf{z})$.

ELBO Optimization

ELBO

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- ▶ ϕ denotes the parameters of the variational posterior $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- ▶ θ represents the parameters of the generative model $p_{\theta}(\mathbf{x}|\mathbf{z})$.

The remaining step is to obtain **unbiased** Monte Carlo estimates of the gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$ and $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$.

Outline

1. Variational Evidence Lower Bound (ELBO)
2. Amortized Inference
3. ELBO Gradients, Reparametrization Trick
4. Variational Autoencoder (VAE)

ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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Gradient $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\theta} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z}$$

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ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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$$\begin{aligned}\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \nabla_{\theta} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\ &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \\ &\approx \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\phi}(\mathbf{z}|\mathbf{x}).\end{aligned}$$

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Naive Monte Carlo Estimation

$$\log p_{\theta}(\mathbf{x}) \geq \int \log p_{\theta}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x}|\mathbf{z}_k), \quad \mathbf{z}_k \sim p(\mathbf{z}).$$

ELBO Gradients: $\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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The variational posterior $q_{\phi}(\mathbf{z}|\mathbf{x})$ typically concentrates more probability mass in a much smaller region than the prior $p(\mathbf{z})$.

ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

Gradient $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

Unlike the θ -gradient, the density $q_{\phi}(\mathbf{z}|\mathbf{x})$ now depends on ϕ , so standard Monte Carlo estimation can't be applied:

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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Reparametrization Trick (LOTUS Trick)

Assume $\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})$ is generated by a random variable $\epsilon \sim p(\epsilon)$ via a deterministic mapping $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$. Then,

$$\mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \mathbf{f}(\mathbf{z}) = \mathbb{E}_{\epsilon \sim p(\epsilon)} \mathbf{f}(\mathbf{g}_{\phi}(\mathbf{x}, \epsilon))$$

ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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Note: The LHS expectation is with respect to the parametric distribution $q_{\phi}(\mathbf{z}|\mathbf{x})$, while the RHS is for the non-parametric $p(\epsilon)$.

ELBO Gradients: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

Reparametrization Trick (LOTUS Trick)

$$\nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \mathbf{f}(\mathbf{z}) d\mathbf{z} = \nabla_{\phi} \int p(\epsilon) \mathbf{f}(\mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon$$

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where $\epsilon^* \sim p(\epsilon)$.

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where $\epsilon^* \sim p(\epsilon)$.

Variational Assumption

$$p(\epsilon) = \mathcal{N}(0, \mathbf{I}); \quad \mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon) = \sigma_{\phi}(\mathbf{x}) \odot \epsilon + \mu_{\phi}(\mathbf{x});$$

$$q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})).$$

Here, $\mu_{\phi}(\cdot)$ and $\sigma_{\phi}(\cdot)$ are parameterized functions (outputs of a neural network).

Thus, we can write $q_{\phi}(\mathbf{z}|\mathbf{x}) = \text{NN}_{e, \phi}(\mathbf{x})$, the **encoder**.

ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

$$\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

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Reconstruction Term

$$\begin{aligned} \nabla_{\phi} \int q_{\phi}(\mathbf{z}|\mathbf{x}) \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} &= \int p(\epsilon) \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon \approx \\ &\approx \nabla_{\phi} \log p_{\theta}(\mathbf{x} | \sigma_{\phi}(\mathbf{x}) \odot \epsilon^* + \mu_{\phi}(\mathbf{x})), \quad \text{where } \epsilon^* \sim \mathcal{N}(0, \mathbf{I}) \end{aligned}$$

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The generative distribution $p_{\theta}(\mathbf{x}|\mathbf{z})$ can be implemented as a neural network.

We may write $p_{\theta}(\mathbf{x}|\mathbf{z}) = \text{NN}_{d, \theta}(\mathbf{z})$, called the **decoder**.

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KL Term

$p(\mathbf{z})$ is the prior over latents $\mathbf{z}$, typically $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.

$$\nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) = \nabla_{\phi} D_{\text{KL}}(\mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})) \| \mathcal{N}(0, \mathbf{I}))$$

ELBO Gradient: $\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x})$

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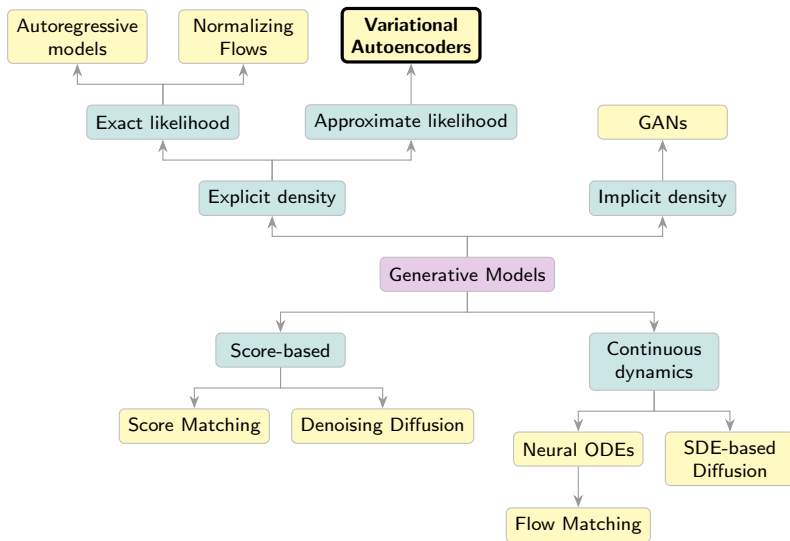
$$\nabla_{\phi} D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) = \nabla_{\phi} D_{\text{KL}}(\mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})) \| \mathcal{N}(0, \mathbf{I}))$$

This expression admits a closed-form analytic solution.

Outline

1. Variational Evidence Lower Bound (ELBO)
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Generative Models Taxonomy



Variational Autoencoder (VAE)

Training

- ▶ Pick a batch of samples $\{\mathbf{x}_i\}_{i=1}^B$ (here we use Monte Carlo technique).

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Note: The encoder $q_\phi(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Variational Autoencoder

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

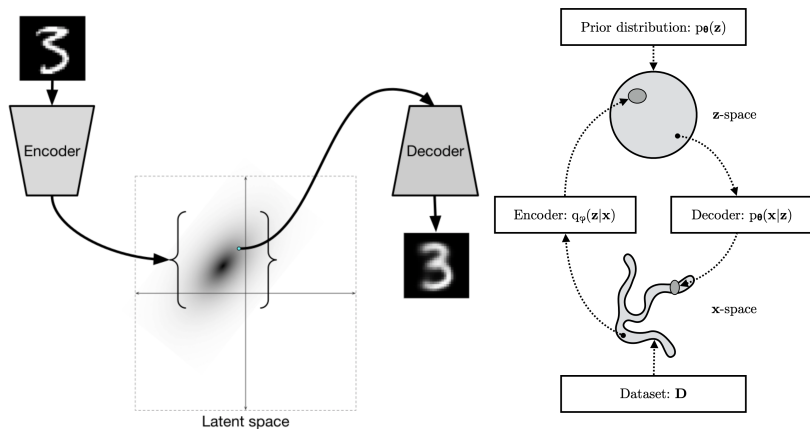
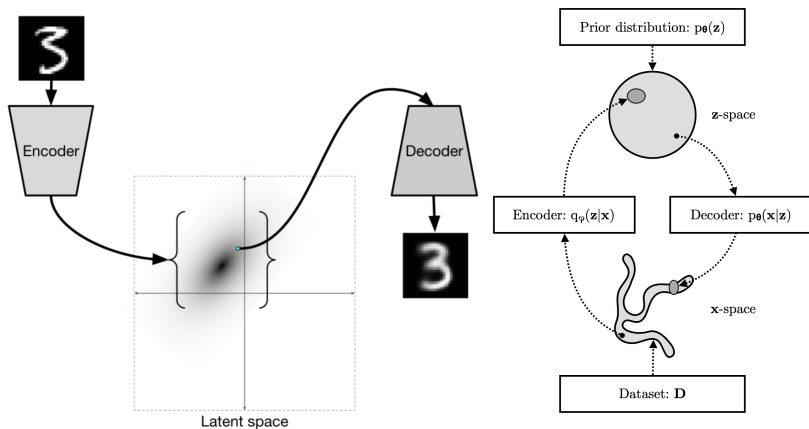


image credit: <http://ijdykeman.github.io/ml/2016/12/21/cvae.html>
Kingma D. P., Welling M., *An Introduction to Variational Autoencoders*, 2019

Variational Autoencoder

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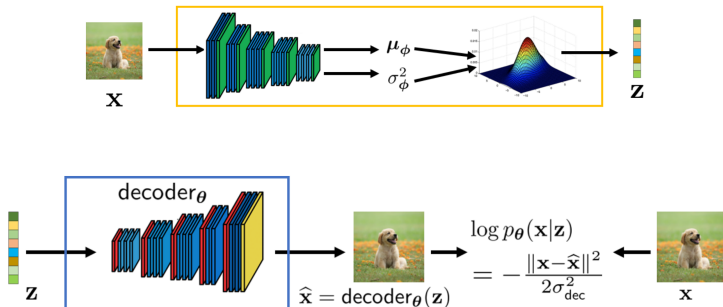


VAEs are widely used as a preliminary stage of projecting data onto low-dimensional space.

image credit: <http://ijdykeman.github.io/ml/2016/12/21/cvae.html>
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Variational Autoencoder

- ▶ The encoder $q_\phi(\mathbf{z}|\mathbf{x}) = \text{NN}_{e,\phi}(\mathbf{x})$ outputs $\mu_\phi(\mathbf{x})$ and $\sigma_\phi(\mathbf{x})$.
- ▶ The decoder $p_\theta(\mathbf{x}|\mathbf{z}) = \text{NN}_{d,\theta}(\mathbf{z})$ outputs parameters of the observed data distribution.



VAE vs Normalizing Flows

	VAE	NF
Objective	ELBO $\mathcal{L}$	Forward KL/MLE
Encoder	stochastic $\mathbf{z} \sim q_{\phi}(\mathbf{z} \mathbf{x})$	deterministic $\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x})$ $q_{\theta}(\mathbf{z} \mathbf{x}) = \delta(\mathbf{z} - \mathbf{f}_{\theta}(\mathbf{x}))$
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Theorem

MLE for a normalizing flow is equivalent to maximizing the ELBO for a VAE where:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{f}_{\theta}^{-1}(\mathbf{z})); \quad q_{\theta}(\mathbf{z}|\mathbf{x}) = \delta(\mathbf{z} - \mathbf{f}_{\theta}(\mathbf{x})).$$

Summary

- ▶ LVMs maximize the variational evidence lower bound (ELBO) to obtain maximum likelihood estimates for the parameters.
- ▶ Parametric posterior distribution $q_{\phi}(\mathbf{z}|\mathbf{x})$ makes the method scalable.
- ▶ The reparametrization trick provides unbiased gradients with respect to the variational posterior $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- ▶ The VAE model is a latent variable model parameterized by two neural networks: a stochastic encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ and a stochastic decoder $p_{\theta}(\mathbf{x}|\mathbf{z})$.
- ▶ Nowadays, the main role of VAEs is to project data into low-dimensional latent space.